

Abhinav Komanduri

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abhinavkomanduri.github.io

EDUCATION

University of Washington

Seattle, Washington

Ph.D., Electrical & Computer Engineering

2026 - Present

- Advised by Dr. Shwetak Patel

University of Arkansas

Fayetteville, Arkansas

B.S., Electrical Engineering

2022 - 2026

- GPA: 3.94/4.0
- Minor in Mathematics
- Thesis: *Design and Development of an eTextile using Mutual Capacitive and Inertial Sensor Fusion for Upper-Limb Posture Estimation*

RESEARCH EXPERIENCE

Applied Embedded Systems and IoT Research Lab (ÆSIR)

Fayetteville, AR

Undergraduate Research Assistant

Aug 2023 - Jun 2026

- Principal Investigator: Alexander H. Nelson
- Conducted research in embedded systems and hardware security. Focus areas include capacitive sensor arrays, fault injection on SRAM PUFs, and secure microcontroller architectures.
- Designed and tested custom PCBs, implemented embedded C firmware across MSP430 and ESP32 platforms, and utilized FPGA-based acceleration for real-time data processing.
- Contributed to peer-reviewed conference publications; awarded funding through the Honors College Research Grant and Goldwater Scholarship.

Inan Research Lab

Atlanta, GA

SURE REU, Georgia Institute of Technology

May 2025 - Jul 2025

- Principal Investigator: Omer T. Inan
- Worked to develop deep learning models for prediction of blood volume decomposition status (BVDS) based on ECG, SCG, and PPG signals from wearable sensing device on pig hearts.
- Read scientific literature, understood the context of a research question, used a deep learning framework to code, train, validate, and test an AI model with minor supervision, and interpreted and presented my results in an oral and poster presentation.
- Attended seminars and won 1st prize at final research symposium.

PEER-REVIEWED PUBLICATIONS

Abhinav Komanduri and Alexander H. Nelson. Power PUFs: Strengthening SRAM PUFs Against Fault Injection on Low-Cost IoT Devices. *2025 IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events (PerCom Workshops)*. 2025.

Abhinav Komanduri and Alexander H. Nelson. Daisy-Chaining Embedded Processors for Enhanced Capacitive Sensor Array Resolution. *2024 IEEE Green Technologies Conference (Green-Tech)*. 2024.

PRESENTATIONS AND POSTERS

📄 “Predicting Hemodynamic Blood Volume Loss based on Multi-Channel Physiological Signals using Deep Learning“, 2025 Georgia Tech SURE Research Symposium, Atlanta, GA, July 2025. (presentation + poster)

“Power PUFs: Strengthening SRAM PUFs Against Fault Injection on Low-Cost IoT Devices“, 2025 IEEE International Conference on Pervasive Computing and Communications (PerCom), Washington, D.C. March 2025. (poster)

“Daisy-Chaining Embedded Processors for Enhanced Capacitive Sensor Array Resolution“, 2024 IEEE Green Technologies Conference (GreenTech), Springdale, AR. April 2024. (poster)

“An Analysis of a Finite Element Based Framework for Capacitive Sensing Simulations“, 85th Annual Meeting of OK-AR MAA Section, Fayetteville, AR. April 2024. (presentation)

✂“Designing Embedded Systems to Monitor EMFs and Power Grids in Real-Time“, 15th Annual Honors Engineering Symposium, Fayetteville, AR. April 2022. (presentation + poster)

HONORS AND AWARDS

Graduate Research Fellowship (\$159,000) Apr 2026
U.S. National Science Foundation (NSF)

Outstanding Undergraduate Researcher Award (Honorable Mention) Dec 2025
Computing Research Association (CRA)

1st Place Oral Presentation Jul 2025
Georgia Institute of Technology SURE Program

Barry Goldwater Scholarship (\$7,500) Mar 2025
Barry Goldwater Foundation

Honors College Research Grants (2) (\$5,000) Jan 2024/2025
University of Arkansas Honors College

Honors College Fellowship (\$80,000) 2022-2026
University of Arkansas Honors College

Arkansas Governor’s Distinguished Scholarship (\$40,000) 2022-2026
Arkansas Division of Higher Education

PROFESSIONAL EXPERIENCE

Wolfspeed, Inc. Fayetteville, AR
Power Modules Backend Operations Intern May 2024 - Aug 2024

- Worked with Power Modules Test Engineering team to design high reliability modules for electrical testing on Wolfspeed Power Modules to run a Gage R&R study.
- Analyzed past module test data using JMP to compile module failure trends.

University of Arkansas, Mathematical Sciences Fayetteville, AR
Grading Assistant Aug 2023 - Present

- Graded homeworks, quizzes, and exams for College Algebra, Elementary Differential Equations, and Calculus 1. Interacted with course instructors to create rubrics and a fair grading scale